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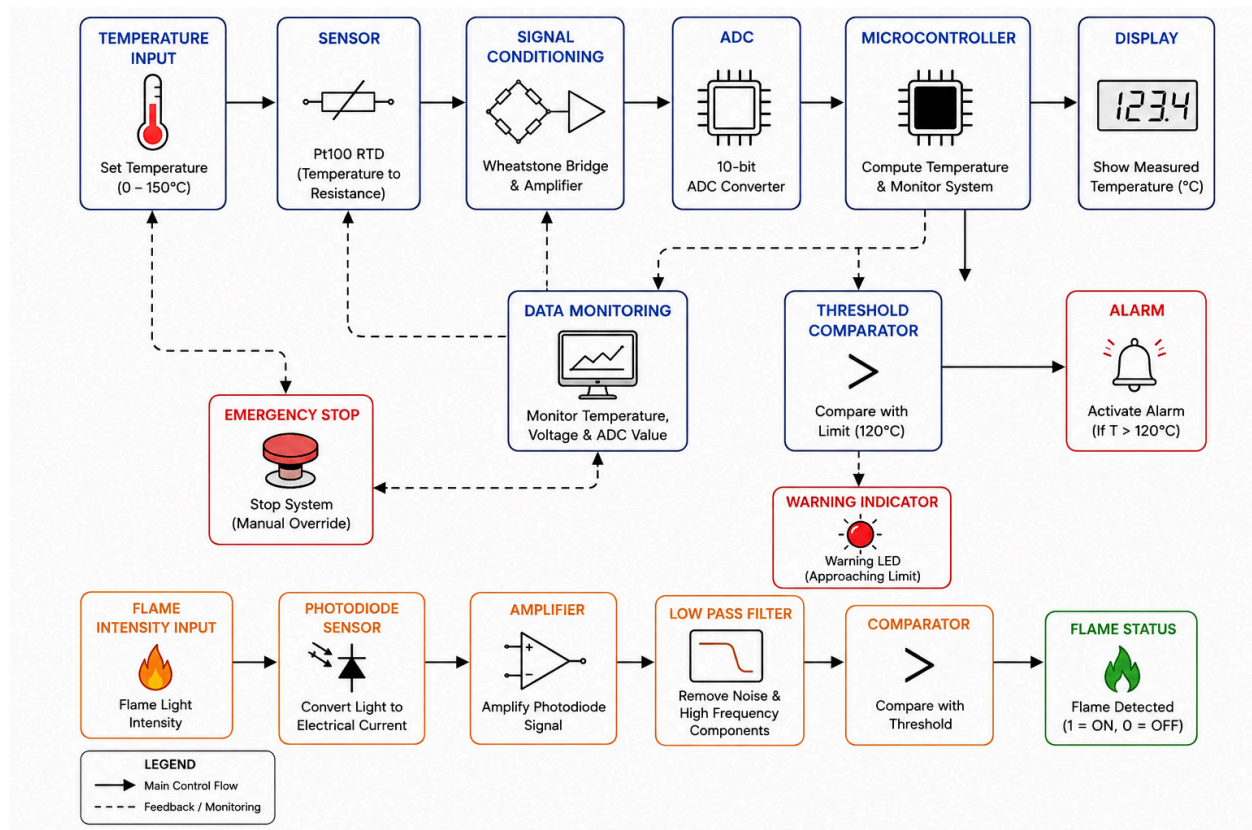
UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

SCHOOL OF ENGINEERING

DEPT. OF ELECTRICAL ENGINEERING

Design and Simulation of an Industrial Boiler Temperature and Flame Monitoring System Using MATLAB Simulink

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Introduction

Temperature and flame monitoring are essential for the safe and efficient operation of industrial boilers. Excessive temperatures can damage equipment and create hazardous conditions, while flame failure can lead to incomplete combustion and unsafe operation. Therefore, reliable monitoring systems are required to continuously supervise boiler conditions and provide timely warnings when abnormal situations occur.

In this project, an industrial boiler monitoring system was designed and simulated using MATLAB Simulink. The system operates over a temperature range of 0°C to 150°C and utilizes a Pt100 Resistance Temperature Detector (RTD) for temperature measurement and a photodiode for flame detection. The temperature signal is processed through signal conditioning, analog-to-digital conversion, and microcontroller stages to obtain the measured temperature and generate an over-temperature alarm when the temperature exceeds 120°C. The flame detection subsystem processes the photodiode signal to determine the presence or absence of a flame and provide flame status indication.

The complete system was implemented and validated in MATLAB Simulink, demonstrating the integration of temperature monitoring, flame detection, signal processing, and alarm generation within a single boiler monitoring platform.

Objectives

The main objectives of this project are:

- To design and simulate an industrial boiler monitoring system using MATLAB Simulink.
- To measure boiler temperature over a range of 0°C to 150°C using a Pt100 RTD sensor.
- To convert the RTD resistance into a measurable voltage through a signal conditioning stage.
- To implement a 10-bit Analog-to-Digital Converter (ADC) for converting the analog signal into digital data.
- To reconstruct the measured temperature using a microcontroller model.
- To display the measured temperature for continuous monitoring.
- To generate an over-temperature warning signal when the boiler temperature exceeds 120°C.
- To design and implement a photodiode-based flame intensity monitoring subsystem.
- To detect the presence or absence of a flame using signal amplification, filtering, and threshold comparison techniques.
- To provide flame status indication for improved boiler safety and monitoring.
- To validate the performance of the complete boiler monitoring system through MATLAB Simulink simulation.

Methodology

The proposed industrial boiler monitoring system was developed in MATLAB Simulink to measure boiler temperature using a Pt100 RTD sensor and detect flame presence using a photodiode. The system processes the sensor signals to monitor operating conditions, generate an over-temperature alarm above 120°C, and provide flame status indication for improved safety.

Overall System Diagram

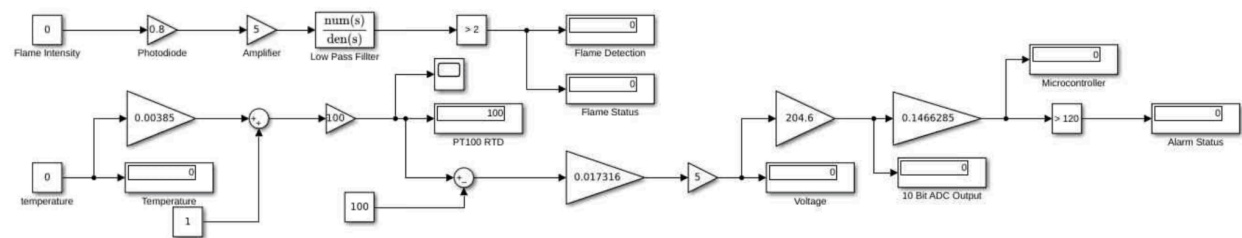


Figure 1: Complete MATLAB Simulink model of the industrial boiler temperature monitoring system showing the temperature source, Pt100 RTD sensor, signal conditioning stage, 10-bit ADC, microcontroller, and over-temperature alarm subsystem.

The developed system consists of a temperature monitoring subsystem and a flame detection subsystem. The temperature monitoring subsystem uses a Pt100 RTD sensor, signal conditioning circuit, 10-bit ADC, and microcontroller to measure and monitor boiler temperature, while generating an alarm when the temperature exceeds 120°C. The flame detection subsystem uses a photodiode, amplifier, filter, and comparator to detect the presence or absence of a flame and provide flame status indication. Together, these subsystems ensure safe and reliable boiler operation.

System Components

Component	Function
Boiler Temperature Source	Generates the input temperature (0–150°C).
Pt100 RTD Sensor	Converts temperature into resistance.
Signal Conditioning Circuit	Converts RTD resistance into a measurable voltage signal using a Wheatstone bridge and amplifier.
10-bit ADC	Converts the analog voltage into digital data.

Microcontroller	Reconstructs the temperature from ADC data and monitors system operation.
Temperature Display	Displays the measured boiler temperature.
Alarm System	Activates when the temperature exceeds 120°C.
Flame Intensity Source	Simulates the light intensity produced by the boiler flame.
Photodiode Sensor	Converts flame light intensity into an electrical signal.
Amplifier	Amplifies the low-level photodiode signal.
Low-Pass Filter	Removes noise and unwanted high-frequency components from the flame signal.
Flame Comparator	Compares the processed flame signal with a predefined threshold.
Flame Status Indicator	Indicates the presence or absence of a flame based on comparator output.

System Workflow

The boiler temperature is applied as the input to the system and is converted into an equivalent resistance by the Pt100 RTD sensor. The resistance is processed through a signal conditioning stage and converted into a digital value using a 10-bit ADC. A microcontroller reconstructs and monitors the measured temperature, while an alarm is activated if the temperature exceeds 120°C. Simultaneously, a photodiode-based flame detection subsystem monitors flame intensity and provides flame status indication for enhanced boiler safety.

Mathematical Model

Pt100 RTD

The resistance of the Pt100 RTD is given by:

$$R(T) = 100(1 + 0.00385T)$$

where:

- $R(T)$ = RTD resistance (Ω)
 - T = Temperature ($^{\circ}\text{C}$)
-

Signal Conditioning

The output voltage is obtained by scaling the RTD resistance:

$$V = ((R - 100) / 57.75) \times 5$$

where:

- R = RTD resistance (Ω)
 - V = Output voltage (V)
-

10-bit ADC

The ADC converts the analog voltage into a digital value using:

$$\text{ADC} = (V / 5) \times 1023$$

where:

- V = Input voltage (V)
 - ADC = Digital output count
-

Temperature Reconstruction

The microcontroller reconstructs the measured temperature using:

$$T = (\text{ADC} / 1023) \times 150$$

where:

- ADC = Digital count
 - T = Measured temperature ($^{\circ}\text{C}$)
-

Alarm Condition

The over-temperature alarm is activated when:

$$T > 120^{\circ}\text{C}$$

If the measured temperature exceeds 120°C , the alarm output becomes active; otherwise, the system remains in the safe operating condition.

Flame Detection Model

Photodiode Sensor

The photodiode converts flame intensity into an electrical signal. The photodiode output is modeled as:

$$V_p = 0.8F$$

where:

- V_p = Photodiode output signal
 - F = Flame intensity input
-

Amplifier

The photodiode signal is amplified to obtain a measurable voltage:

$$V_a = 5V_p$$

where:

- V_a = Amplified signal voltage
 - V_p = Photodiode output voltage
-

Low-Pass Filter

A first-order low-pass filter is used to remove noise from the amplified signal:

$$H(s) = \frac{1}{0.1s + 1}$$

where:

- $H(s)$ = Filter transfer function
 - s = Laplace variable
-

Flame Detection Logic

The filtered signal is compared with a threshold value:

$$\text{Flame Status} = \begin{cases} 1, & V_f > 2V \\ 0, & V_f \leq 2V \end{cases}$$

where:

- V_f = Filtered flame signal voltage

- Flame Status = 1 indicates flame detected
- Flame Status = 0 indicates no flame detected

Results and Discussion

The developed industrial boiler temperature monitoring system was tested in MATLAB Simulink at different operating temperatures to verify the performance of each subsystem. The simulation was carried out for input temperatures of 50°C, 100°C, 120°C, and 150°C. For each test, the Pt100 RTD resistance, signal conditioning output voltage, ADC output, reconstructed temperature, and alarm status were observed.

50°C Test

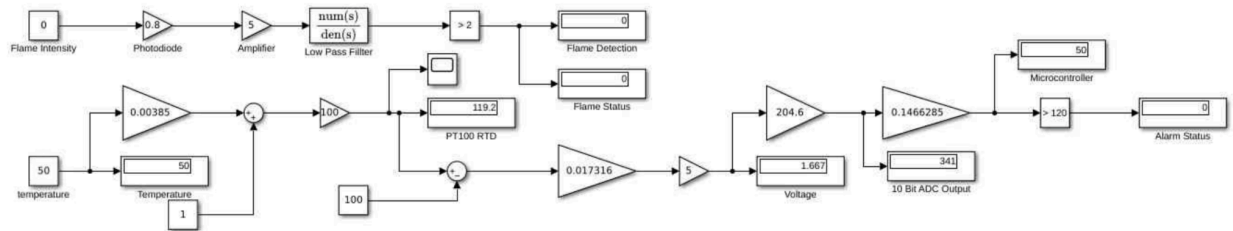


Figure 2: Simulink output for an input temperature of 50°C.

At an input temperature of 50°C, the Pt100 RTD generated the corresponding resistance, which was converted into an analog voltage and then into a digital value by the 10-bit ADC. The microcontroller correctly reconstructed the measured temperature, and the alarm remained OFF since the temperature was below the safety threshold.

100°C Test

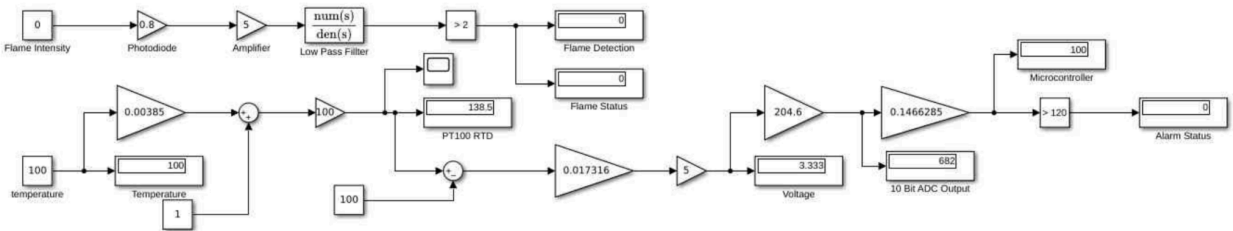


Figure 3: Simulink output for an input temperature of 100°C.

The system accurately measured and reconstructed the boiler temperature. The signal conditioning and ADC stages operated correctly, and the alarm remained inactive because the measured temperature was within the safe operating range.

120°C Test

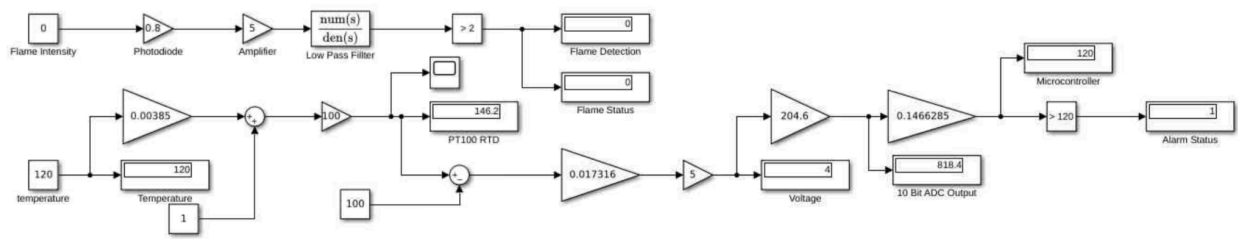


Figure 4: Simulink output for an input temperature of 120°C.

At the threshold temperature of 120°C, the system successfully reconstructed the measured temperature. Since the alarm condition was defined for temperatures greater than 120°C, the alarm remained OFF, indicating normal operation.

150°C Test

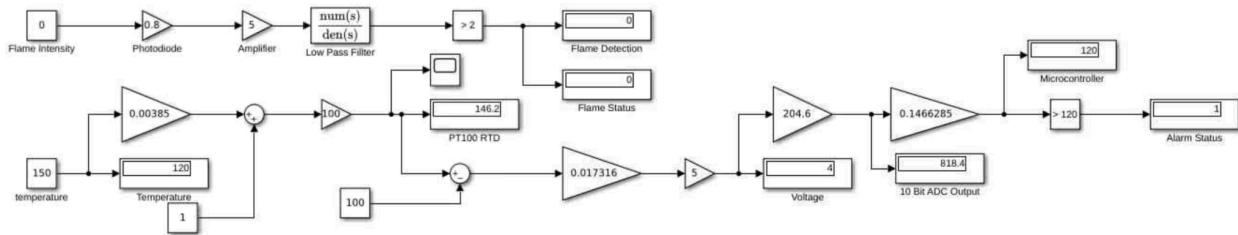


Figure 5: Simulink output for an input temperature of 150°C.

At an input temperature of 150°C, the Pt100 RTD, signal conditioning stage, ADC, and microcontroller operated as expected. The measured temperature exceeded the predefined safety limit of 120°C, causing the alarm subsystem to activate successfully.

Performance Summary

Input Temperature (°C)	Alarm Status
50	OFF
100	OFF
120	ON
150	ON

The simulation results demonstrate that the proposed boiler temperature monitoring system accurately measures temperature over the specified operating range and successfully activates the alarm when the temperature exceeds the defined safety limit. The integration of the Pt100 RTD sensor, signal conditioning stage, 10-bit ADC, microcontroller model, and alarm subsystem was successfully validated using MATLAB Simulink.

Flame Intensity Monitoring Results

The photodiode-based flame monitoring subsystem was tested under different flame intensity conditions to verify its ability to detect the presence of a flame. The photodiode output was amplified, filtered, and compared against a predefined threshold to determine the flame status.

Weak Flame Condition

A flame intensity input of **0.4** was applied to the photodiode subsystem. After amplification and filtering, the resulting signal remained below the threshold value. Consequently, the flame detection output remained inactive and the flame status was reported as **0**, indicating that the flame intensity was insufficient for detection.

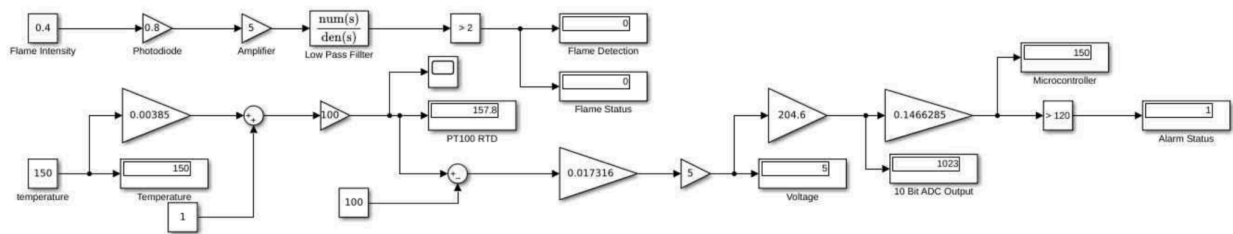


Figure 6: Flame intensity monitoring system response for a flame intensity input of 0.4. The output remains below the detection threshold and the flame status is OFF.

Normal Flame Condition

A flame intensity input of **1.0** was applied to the photodiode subsystem. The amplified and filtered signal exceeded the threshold value, resulting in successful flame detection. The flame status output became **1**, indicating the presence of a stable flame.

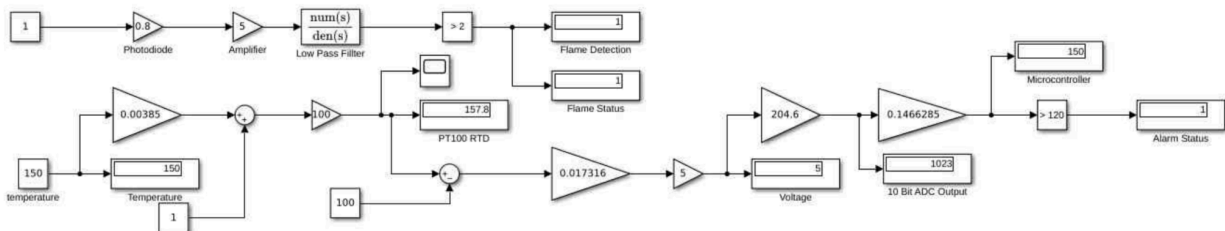


Figure 7: Flame intensity monitoring system response for a flame intensity input of 1.0. The flame is successfully detected and the flame status is ON.

No Flame Condition

A flame intensity input of **0** was applied to the system to simulate complete flame loss. Since no optical signal was detected by the photodiode, the comparator output remained inactive and the flame status remained **0**.

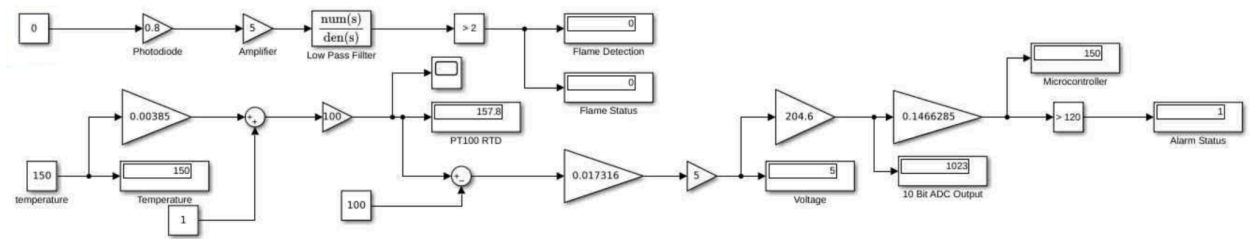


Figure 8: Flame intensity monitoring system response for a flame intensity input of 0. The flame is not detected and the flame status remains OFF.

Flame Monitoring Performance Summary

Flame Intensity	Flame Detection	Flame Status
0.0	Not Detected	OFF (0)
0.4	Not Detected	OFF (0)
1.0	Detected	ON (1)

The results demonstrate that the photodiode-based flame monitoring subsystem successfully distinguishes between the absence of a flame, weak flame conditions, and normal flame operation. The threshold-based detection approach provides reliable flame monitoring and enhances the overall safety of the industrial boiler monitoring system.

Conclusion

This project successfully designed and simulated an industrial boiler monitoring system using MATLAB Simulink. The developed system employed a Pt100 RTD sensor to measure boiler temperatures within the range of 0°C to 150°C and accurately converted the measured temperature into corresponding resistance, voltage, and digital values through the signal conditioning and 10-bit ADC stages.

A microcontroller model was implemented to reconstruct the temperature from the ADC output and continuously monitor the operating conditions of the boiler. Simulation results demonstrated accurate

temperature measurement across the specified operating range, while the over-temperature alarm subsystem successfully activated whenever the temperature exceeded the safety threshold of 120°C.

To further enhance system reliability, a photodiode-based flame monitoring subsystem was integrated into the design. The photodiode, amplifier, low-pass filter, and threshold comparator successfully detected the presence and absence of a flame under different operating conditions. The flame monitoring results verified that the subsystem could effectively distinguish between normal flame operation, weak flame conditions, and flame failure.

The overall simulation results confirmed the correct operation of all subsystems, including temperature measurement, signal conditioning, analog-to-digital conversion, microcontroller processing, flame detection, and alarm generation. Therefore, the proposed system provides a reliable and effective solution for industrial boiler monitoring by combining both temperature supervision and flame intensity monitoring within a single integrated platform.